

An Exact Rational Solution with $y = 9162$

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Abstract

This note records and verifies an exact rational nonintegral solution of a Diophantine equation with two rational parameters. The solution gives $y = 9162$ (and, by symmetry, $y = -9162$). All calculations are performed in exact rational arithmetic. A final remark explains why this solution is valid for the original rational search but does not meet the later, stronger restriction that the reduced denominators of k_1 and k_2 divide the two large coefficients.

1 The equation

Set

$$\begin{aligned} A &= 7729484335457653901640057298531371241781, \\ B &= 7668575607239450973459863267707132263860, \\ C &= 2486598372481845396683104279916570951657, \\ D &= 609530524018264138310326718615033307496. \end{aligned}$$

We consider

$$y^2 = (6(Ak_1 + B) + (Ck_2 + D))^2 + \frac{36(Ak_1 + B)^3 - 19}{Ck_2 + D}. \quad (1)$$

It is convenient to introduce

$$n = Ak_1 + B, \quad x = Ck_2 + D. \quad (2)$$

Then (1) becomes

$$y^2 = (6n + x)^2 + \frac{36n^3 - 19}{x}. \quad (3)$$

2 The rational parameter values

Take

$$n = -\frac{1506}{5}, \quad x = -\frac{61}{5}. \quad (4)$$

Solving (2) for k_1, k_2 gives

$$k_1 = \frac{n - B}{A}, \quad k_2 = \frac{x - D}{C}. \quad (5)$$

Substituting (4) into (5) yields

$$k_1 = -\frac{38342878036197254867299316338535661320806}{38647421677288269508200286492656856208905} \quad (6)$$

and

$$k_2 = -\frac{3047652620091320691551633593075166537541}{12432991862409226983415521399582854758285}. \quad (7)$$

Indeed, the numerators and denominators arise from

$$\begin{aligned} 5B + 1506 &= 38342878036197254867299316338535661320806, \\ 5A &= 38647421677288269508200286492656856208905, \\ 5D + 61 &= 3047652620091320691551633593075166537541, \\ 5C &= 12432991862409226983415521399582854758285. \end{aligned}$$

Moreover,

$$\begin{aligned} \gcd(5B + 1506, 5A) &= 1, \\ \gcd(5D + 61, 5C) &= 1. \end{aligned}$$

Thus the fractions in (6) and (7) are reduced, and both k_1 and k_2 are rational nonintegers.

3 Exact verification of the affine substitutions

Using (6),

$$\begin{aligned} Ak_1 + B &= A \left(-\frac{5B + 1506}{5A} \right) + B \\ &= -\frac{5B + 1506}{5} + B \\ &= -\frac{1506}{5}. \end{aligned}$$

Similarly, using (7),

$$\begin{aligned} Ck_2 + D &= C \left(-\frac{5D + 61}{5C} \right) + D \\ &= -\frac{5D + 61}{5} + D \\ &= -\frac{61}{5}. \end{aligned}$$

Consequently, the large-parameter equation (1) is reduced exactly to (3) with the values in (4).

4 Exact verification that $y = 9162$

First,

$$6n + x = 6 \left(-\frac{1506}{5} \right) - \frac{61}{5} = -\frac{9097}{5}, \quad (8)$$

so

$$(6n + x)^2 = \frac{9097^2}{25} = \frac{82755409}{25}. \quad (9)$$

Next,

$$\begin{aligned} 36n^3 - 19 &= 36 \left(-\frac{1506}{5} \right)^3 - 19 \\ &= -\frac{122963842151}{125}. \end{aligned}$$

Since $x = -61/5$, it follows that

$$\begin{aligned}\frac{36n^3 - 19}{x} &= \left(-\frac{122963842151}{125}\right) \left(-\frac{5}{61}\right) \\ &= \frac{2015800691}{25}.\end{aligned}\tag{10}$$

Combining (9) and (10),

$$\begin{aligned}(6n + x)^2 + \frac{36n^3 - 19}{x} &= \frac{82755409 + 2015800691}{25} \\ &= \frac{2098556100}{25} \\ &= 83942244 \\ &= 9162^2.\end{aligned}$$

Therefore the exact solutions for y are

$$\boxed{y = 9162 \quad \text{or} \quad y = -9162.}\tag{11}$$

In particular, the requested positive value $y = 9162$ is valid.

5 Status under the later denominator restriction

The reduced denominators of the two rational parameters are

$$\begin{aligned}\text{den}(k_1) &= 5A = 38647421677288269508200286492656856208905, \\ \text{den}(k_2) &= 5C = 12432991862409226983415521399582854758285.\end{aligned}$$

Hence this solution is a valid solution of the original rational-parameter problem, but it does *not* satisfy the later conditions

$$\text{den}(k_1) \mid A, \quad \text{den}(k_2) \mid C.$$

The obstruction is exactly the extra factor 5, which is also the common denominator of $n = -1506/5$ and $x = -61/5$.

A Exact Magma verification

The following Magma code verifies the solution without floating-point arithmetic.

```
Q := Rationals();

A := 7729484335457653901640057298531371241781;
B := 7668575607239450973459863267707132263860;
C := 2486598372481845396683104279916570951657;
D := 609530524018264138310326718615033307496;

k1 := Q!(-38342878036197254867299316338535661320806)
      / 38647421677288269508200286492656856208905;

k2 := Q!(-3047652620091320691551633593075166537541)
      / 12432991862409226983415521399582854758285;

y := 9162;

n := A*k1 + B;
x := C*k2 + D;

assert n eq Q!(-1506)/5;
assert x eq Q!(-61)/5;
assert y^2 eq (6*n + x)^2 + (36*n^3 - 19)/x;

assert GCD(Abs(Numerator(k1)), Denominator(k1)) eq 1;
assert GCD(Abs(Numerator(k2)), Denominator(k2)) eq 1;

print "n =", n;
print "x =", x;
print "y =", y;
print "RHS =", (6*n + x)^2 + (36*n^3 - 19)/x;
```